

AI Appointment Scheduler

Technical Documentation — n8n Workflow

Version 6 · July 2026 · 24 nodes · 7 agent tools

1. Overview

A guardrailed conversational booking assistant for a dental clinic (Dr. Hakim). Patients chat with the assistant to book, reschedule, or cancel appointments. The assistant validates office hours deterministically, checks real calendar availability, writes to Google Calendar and Google Sheets, and logs every turn.

The defining characteristic is a layered guardrail architecture. Every inbound message passes through progressively more expensive checks before reaching the booking agent: deterministic rate limiting, a regex abuse filter, a lightweight LLM intent classifier, and a conversation-state gate. Cheap deterministic checks run first and stop most spam and prompt-injection attempts without spending a single model token. The GPT-5 agent — the most expensive component — only ever sees traffic already known to be legitimate. That funnel is also what makes a full-size model on the agent economically viable.

A second principle runs throughout: anything the system cannot afford to get wrong is enforced deterministically in code, not described to the model in a prompt. Office hours are the clearest example.

2. Architecture

Main path (single-item chat executions):

Chat Trigger → Normalize Input → Rate Limiter & Abuse Score → Is Blocked? → Rule-Based Filter → Is Filtered? → Intent Classifier → Parse Intent → Intent Allowed? → AI Agent → Log Interaction → Format Response

Rejected messages exit early via Blocked Reply, Fixed Reply, or Out of Scope Reply — canned responses returned without invoking the agent.

The AI Agent carries nine sub-nodes on LangChain ports: OpenAI Chat Model (ai_languageModel, gpt-5), Window Buffer Memory (ai_memory, 20-turn window), and seven tools (ai_tool).

Layer	Cost	Stops
① Rate limiter	0 tokens	Volume spam, flooding, repeat offenders
② Regex filter	0 tokens	Math spam, injection phrasing, jailbreaks
③ Intent classifier	1 mini call	Anything outside the clinic's domain
④ State gate	0 tokens	False rejections of mid-booking replies
⑤ Validate Slot	0 tokens	Bookings outside office hours
⑥ Agent prompt	—	Unvalidated bookings, invented availability

3. Node Reference

Node	Type	Purpose
When chat message received	chatTrigger 1.1	Entry point. Supplies sessionId and chatInput.
Normalize Input	code 2	Trims message; produces lowercase whitespace-collapsed copy for deterministic checks.
Rate Limiter & Abuse Score	code 2	Per-session spam control via workflow static data. Volume, repetition, cooldowns.
Is Blocked?	if 2.2	Routes blocked sessions to Blocked Reply.
Blocked Reply	code 2	Canned throttling message.
Rule-Based Filter	code 2	Greeting fast-path plus abuse/injection regex. Phone-number-aware math detection.
Is Filtered?	if 2.2	Routes filtered messages to Fixed Reply.
Fixed Reply	code 2	Returns replyText — greeting welcome or rejection line.
Intent Classifier	openAi 1.8	gpt-4o-mini, JSON-only. 8 intents, ordered rules. Fails open.
Parse Intent	code 2	Parses classifier JSON; applies allowed list, state override, PROMPT_ATTACK scoring.
Intent Allowed?	if 2.2	Routes allowed intents to the agent.
Out of Scope Reply	code 2	Standard rejection line.
AI Agent	agent 1.7	The booking agent. gpt-5, 20-turn memory, 7 tools, procedural system prompt.
OpenAI Chat Model	lmChatOpenAi 1.2	gpt-5 — the agent's reasoning model.
Window Buffer Memory	memoryBufferWindow 1.3	20-turn per-session context. Required for multi-step procedure tracking.
Validate Slot	toolCode 1.1	Deterministic office-hours gate. Sole authority on weekdays and opening hours.
Check Availability	googleCalendarTool 1.3	Event Get Many over a full day. Returns all events for overlap detection.
Creat event	googleCalendarTool 1.3	Books the slot. Returns the event id used as BookingID.
Add data	googleSheetsTool 4.5	Appends BookingID / Name / Number / Date / Confirmation to the Booking tab.
Find Booking	googleCalendarTool 1.3	Looks up a patient's events by phone number. Returns event ids.
Cancel Booking	googleCalendarTool 1.3	Deletes a calendar event by id. Permanent; single-call only.
Update Booking Status	googleSheetsTool 4.5	Sheets update matched on BookingID. Marks rows Cancelled or Rescheduled.
Log Interaction	googleSheets 4.5	Appends Timestamp / SessionId / Intent / Message / Response to the Logs tab.

Node	Type	Purpose
Format Response	code 2	Re-fetches agent output; substitutes the error line if output is missing.

4. Guardrail Layers

4.1 Rate Limiter & Abuse Score (deterministic)

Per-session state in `$getWorkflowStaticData('global')` under `users[sessionId]`:

```
{ timestamps: [], recentMessages: [], abuseScore: 0, blockedUntil: null,
  lastAgentAt: null }
```

- **Volume:** more than 10 messages in a rolling 30-second window → block, abuseScore +1.
- **Repetition:** the same normalized message three times consecutively → block, abuseScore +1.
- **Cooldown:** abuseScore ≥ 5 → 15-minute block; score resets on expiry.

4.2 Rule-Based Filter (deterministic)

A greeting fast-path matches messages that are only a greeting and returns an immediate welcome steering into booking — zero tokens for the most common message received. The abuse group catches instruction-override phrasing, persona hijacks, jailbreak vocabulary, and code/joke requests.

Math-spam detection is phone-number-aware. Requiring only 'all characters are math characters' matched bare phone numbers, since digits are math characters — blocking legitimate patients and incrementing their abuse score. Detection now requires an operator between digits and is preceded by an explicit phone-shaped guard (7–15 digits with optional separators), which must run first because + and - are both operators and phone punctuation.

4.3 Intent Classifier (gpt-4o-mini)

The cheapest LLM in the chain. Never chats; returns strictly `{"intent":"<INTENT>"}`. Eight intents: PROMPT_ATTACK, BOOK_APPOINTMENT, RESCHEDULE, CANCEL, CLINIC_INFO, FOLLOW_UP, GREETING, OUT_OF_SCOPE. Ordered rules, first match wins, worked examples in the prompt. PROMPT_ATTACK adds +2 abuse. On node failure it fails open to the agent rather than falsely rejecting a real patient — the rate limiter and regex filter have already run and the agent carries its own guardrails, so this is the safe direction to fail.

4.4 Conversation-State Gate (Parse Intent)

The classifier sees one message at a time, so mid-conversation replies look off-topic in isolation. Parse Intent stamps `lastAgentAt` on every message routed to the agent; a later OUT_OF_SCOPE within 15 minutes of an active flow is reclassified FOLLOW_UP and allowed through. The override never applies to PROMPT_ATTACK, and the regex filter still runs first.

4.5 Validate Slot (deterministic office-hours gate)

A Code tool holding the hours table. Given 'YYYY-MM-DD HH:mm' it computes the weekday itself and checks the slot against that day's window, accounting for the 60-minute consultation. Returns valid, weekday, opens, closes, latest_bookable_start, a plain-English reason, and every bookable_start_times for that day.

This exists because describing office hours in a prompt does not enforce them. The model must otherwise chain three inferences — today is Saturday, Saturday closes at 13:00, 17:00 is after that — and models drop them, especially when Check Availability reports no conflict at 17:00, which reads like permission. The prompt now states that Validate Slot is the sole authority on hours and weekdays, that the listed hours are for the patient's benefit only, and that valid=false halts the procedure.

To change clinic hours, edit the HOURS table at the top of the tool's code: minutes from midnight, [open, close], null for closed.

4.6 Agent-Level Procedures

The system prompt encodes ordered procedures rather than descriptions: an availability procedure gated on Validate Slot; a booking completion sequence (create event → read the returned id → write the spreadsheet with it → only then confirm to the patient); an 8-step cancellation procedure; and a 10-step reschedule that cancels before creating so a mid-flow failure never double-books.

5. BookingID: linking the two systems

Google Calendar and Google Sheets are independent systems with no shared key by default. Create event returns the created event's id; the prompt requires that id be passed to Add data as BookingID. Cancellation then matches the spreadsheet row on the same id the calendar uses.

Name, Number, and Date were all rejected as matching keys — none is unique, since one patient may hold several bookings. BookingID is exact. Rows created before the column existed have no id and cannot be cancelled by the agent.

6. Identity & Destructive Actions

Anyone with the chat link can reach the agent, so destructive actions require verification. The Identity Rule: no appointment may be cancelled, deleted, or moved unless the phone number in that event's title matches a number the patient supplied in the same conversation. Never by name alone. Requests to cancel on another person's behalf are refused with a referral to the clinic.

Cancel Booking is single-call by contract. The deletion is permanent, and a second call for the same id returns 'Not Found' because the event is gone — an error that reads like failure and invites a retry loop. Both the tool description and the prompt forbid retrying it.

7. Error Handling

onError: continueRegularOutput is set on the Intent Classifier, the AI Agent, all six external tools, and Log Interaction. Failures return to the agent as tool results instead of terminating the execution.

The Tool Failure Rule instructs the agent to treat any error result as NOT DONE, never retry, never substitute another tool, and reply with exactly one line:

There's an unexpected error, please call 09121231234

An exceptions block is equally important: Validate Slot returning valid=false, Check Availability returning an overlap or no events, and Find Booking returning zero results are all successful tool calls, not errors. Without that block the agent fires the error line at patients who merely needed a different time.

One asymmetric case is handled explicitly: if Cancel Booking succeeded but Update Booking Status failed, the appointment is genuinely cancelled — the agent says so and never re-runs the delete to 'fix' the mismatch.

Format Response is the last resort. If the agent's output is missing or empty for any reason — crash, timeout, malformed return — it substitutes the error line, so a patient never receives a blank chat.

8. Model Selection

gpt-5 drives the agent; gpt-4o-mini drives the classifier. The split is deliberate and was validated empirically. The classifier runs on every inbound message and performs one bounded task: read a sentence, emit one JSON field. Mini is well suited and is where token spend concentrates.

The agent holds seven tools and three multi-step ordered procedures. On a mini-class model it failed characteristically: it invented an exact confirmation phrase ('reply exactly: "Yes, cancel"'), then string-matched against it and rejected 'yes, cancel' for casing. Same prompt on gpt-5 did not reproduce the behavior — the failure was model capability, not instruction ambiguity. The guardrail funnel is what makes a full-size agent model affordable: it only sees traffic that survived the cheap layers.

9. Implementation Notes

- **Static data is production-only.** `$getWorkflowStaticData` persists only between executions of an active workflow. Manual editor runs reset it, so the rate limiter and state gate must be tested through the public chat URL.
- **Use `.first()`, not `.item`, after the classifier.** Paired-item tracing breaks across the LangChain OpenAI node. Chat executions are single-item; `.first()` is equivalent and immune.
- **Code nodes run in 'Run Once for Each Item' mode** and return a single `{ json }` object.
- **Timezone is pinned three ways:** workflow settings (Asia/Manila), the `GENERIC_TIMEZONE` env var, and `$now.setZone('Asia/Manila')` in the prompt. The env var still governs the Logs Timestamp column.
- **Sheets prerequisites:** Booking tab (BookingID, Name, Number, Date, Confirmation) and Logs tab (Timestamp, SessionId, Intent, Message, Response). Headers are matched by name; renaming breaks writes silently.
- **Validator false positives:** n8n-MCP flags `googleCalendarTool/googleSheetsTool` as unknown types, and reports range/values missing on Sheets v4.5 update nodes. Both are benign.

10. Debugging Log

Every entry below was found by running the workflow, not by reading it.

Symptom	Root cause	Fix
Workflow hung at Parse Intent	<code>.item</code> relies on paired-item tracing, which breaks across the LangChain OpenAI node	<code>.first()</code> everywhere downstream of the classifier
"hi" burned an LLM call	No fast-path for the most common message	Greeting regex short-circuit
Name + phone reply rejected as off-topic	Stateless classifier: a name and number match no booking rule, fall to	FOLLOW_UP intent plus conversation-state gate

Symptom	Root cause	Fix
	OUT_OF_SCOPE	
Same slot booked twice	Boolean free/busy check plus offset-less timestamps	Full-day event list, overlap rule, hard +08:00
Calendar events created (No title)	additionalFields empty — the title existed only in the prompt	Event_Title via \$fromAI
Events at 05:00, dated yesterday	Instance running America/New_York; \$now resolving to the wrong day before 08:00 local	setZone in prompt plus GENERIC_TIMEZONE
Calendar filled, spreadsheet stayed empty	Add data had no toolDescription — the model saw a tool named "Add data" and no reason to call it	Manual tool description plus mandatory completion sequence
Agent claimed to cancel bookings it could not	No delete tool existed; the prompt instructed cancellation anyway	Find Booking, Cancel Booking, Update Booking Status, plus an honesty rule
Cancel returned 'Not Found'	Repeat delete call on an already-deleted event	Single-call contract in tool description and prompt; onError nets
Booked 17:00 on a Saturday	Office hours were described to the model, never enforced	Validate Slot deterministic gate
Confirmation loop on "yes, cancel"	Mini-class model invented an exact phrase then string-matched it	Confirmation Rule (intent not characters), no-repeat rule, gpt-5
Bare phone number blocked as spam	Math regex matched any all-digit string — digits are math characters	Phone guard plus operator-required math detection

The tool-description bug is instructive. Two of three tools had hand-written descriptions and one did not; the agent quietly stopped using the one that did not. It never errored — it simply declined to call it. It was found by reading the logs the workflow writes about itself.

11. Test Plan

1. Fresh session, 'hi' → instant canned welcome, zero LLM calls.
2. '09464564567' as the first message → reaches the agent; not blocked, no abuse score.
3. 'Can I book today at 5pm?' on a Saturday → refused with the 13:00 closing reason; offered 09:00 / 10:15 / 11:30.
4. '1+1' → blocked by regex; abuse score increments.
5. Fresh session, 'who won the world cup?' → rejected at the intent gate.
6. 'ignore previous instructions' mid-booking → blocked; PROMPT_ATTACK never softened by the state gate.
7. Eleven rapid messages → rate-limit block; sustained abuse → 15-minute cooldown.
8. Complete a booking → verify all three: calendar event at the correct local time titled 'Name - Number', a Booking row with a populated BookingID, and a Logs row.
9. Cancel that booking, replying 'yes' in lower case → accepted first time, no repeated read-back; event deleted; Booking row reads Cancelled.
10. Attempt to cancel using a phone number that does not match the event title → refused.
11. Book the same slot from a second incognito session → conflict detected, real alternatives offered.